Build Status — What Is Done and What Is Not

As of 20 September 2026, end of session. Counts in this document are generated from the repository, not maintained by hand: the block results come from `sim/run_sim.py`, the unknowns from the specification's own register.

At a glance

	Count
Circuit blocks simulated	**24**
Blocks passing every check	**24**
Blocks failing a check	**0**
Research memos	**12**
Documents in the bible	**18**
Unknowns closed	**4**
Unknowns partly answered	**4**
Unknowns still open	**9**
Phase	**2** — schematic capture **complete**: 10 sheets, root wired, project netlist checked

All 24 blocks pass. On 19 September it was 16, after a temperature-corner pass found six blocks that fail cold or hot and turned prose caveats into enforced checks. All six are now answered, and the answers are part specifications rather than widened checks — every one of them is written up in `docs/bom_requirements.md` with the block that demands it, and every failing corner is kept as pinned evidence of *why* the requirement exists.

One of the six was not a part-selection problem at all. The metering unit's coil is copper, copper moves 0.39 %/°C, and no better resistor exists to buy. It needed closed-loop current control, which put a 50 mΩ shunt and a sense amplifier on ECU pin 88's return — new schematic content, not a BOM line. See §1.7.

Schematic complete; the PCB is prepared for hand routing, with the housing and connector as placeholders (14 October). Firmware runs on the PC against a simulated engine. The KiCad project is in `hw/`, with all ten hierarchical sheets created and the first — `power_input` — drawn and netlist-checked. See §7.

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1. Circuit blocks — simulated and passing

Each block is a SPICE netlist plus a set of falsifiable checks. A block "passes" when every claim it makes about itself holds. These are design verification, not layout.

1.1 Power chain — 9 blocks

Block	What it covers	Status
`emi_filter`	Two-stage LC, CISPR 25 conducted, differential mode	Passing
`feed_emi`	VBAT_INJ / VBAT_EGR feed filters, CISPR 25 Class 5 against modelled load harmonics	Passing
`reverse_battery`	Ideal-diode P-FET vs series Schottky	Passing
`transient_clamp`	ISO 7637-2 pulse 2a, +112 V / 2 Ω / 50 µs	Passing
`load_dump`	ISO 7637-2 pulse 5b, 40 V / 0.5 Ω / 400 ms	**FAILS — deliberately**
`buck_preregulator`	6–40 V to 5 V at 400 kHz	Passing
`sensor_rail`	5 V sensor distribution, one PTC per group	**FAILS — cold corner**
`mcu_pdn`	3V3 decoupling impedance, 100 mΩ target	**FAILS — cold corner**
`supervisor`	Watchdog, brownout, fail-safe gate kill path	Passing
`negative_pulses`	ISO 7637-2 pulses 1 and 3a, negative excursions	**FAILS — real gap**

Why `load_dump` fails, and why that is correct. The TVS chosen against pulse 2a absorbs 46 J at 116 W under pulse 5b — roughly 100× both its single-pulse energy rating and its continuous dissipation rating. The failure is the deliverable: it says the part selection is wrong, not that the simulation is. The fix is a higher standoff voltage so the TVS stays off during a normal clamped load dump, rather than a larger part to absorb it. ~~Blocked on U16~~ — closed 9 Oct 2026: the owner chose to design for the suppressed (clamped) load dump, so pulse 5b applies.

1.2 Sensor front-ends — 7 blocks

Block	What it covers	Status
`sensor_ratiometric`	Pins 41/35/80/37, 0.5–4.5 V into a 3.3 V ADC	Passing
`sensor_differential`	Shared sensor ground on pin 34, single-ended vs differential	Passing
`ntc_frontend`	Pin 79 charge-air temperature, divider vs current source	Passing
`battery_sense`	Pins 04/06, 6–40 V rail into a 3.3 V ADC	Passing
`discrete_input`	Pins 20/24/71, active-high switched battery	Passing
`trip_module_sense`	Trip-module state, supervised three-state loop	Passing
`cam_frontend`	Pin 46 cam Hall, 5 V swing divided into the 3.3 V domain	Passing

`ntc_frontend` simulates **both** candidate topologies because **U2** is unresolved. One resistance measurement at the machine picks between them; the front-end is designed so that measurement selects a populate option rather than unblocking a redesign.

1.3 Speed and position — 1 block

Block	What it covers	Status
`vr_conditioner`	Crank VR sensor, pins 52/74/30	Passing

1.4 Injection — 3 blocks

Block	What it covers	Status
`injector_boost`	Why the boost stage exists, pins 03/05	**FAILS — both corners**
`injector_turnoff`	Turn-off recirculation into the boost rail	Passing
`boost_converter`	Injector rail reservoir sizing	Passing

`boost_converter` passes but carries a recorded gap: it has **no bleed path.** Its charge source regulates the rail from below only. Harmless while current flowed outward; `injector_turnoff` now pushes energy back in, and a hold-current cutoff would climb the rail about 2 V per event with no way down.

1.5 Actuators — 3 blocks

Block	What it covers	Status
`metering_unit_pwm`	Pin 88, low-side PWM into the fuel metering solenoid	**FAILS — both corners**
`relay_driver`	Pins 50/69, low-side FET with flyback	Passing
`egr_hbridge`	Pins 59/81, positional actuator, feedback on 37	**FAILS — shoot-through**

1.6 Communications — 1 block

Block	What it covers	Status
`can_termination`	Split vs single termination, J1939 250 kbit/s	**FAILS — tempco**

1.7 The six corners that were failing, and what each one bought

All six are closed. None was closed by widening a check: each is answered by specifying a part, with the failing corner kept as pinned evidence. The requirements live in `docs/bom_requirements.md` under the tags below.

Block	Was	Answer	Tag
`can_termination`	123.95 Ω at 200 ppm/°C vs J1939's own 120 Ω	thin-film, 0.1 %, 50 ppm/°C → 121.10 Ω at stacked worst case	`CAN-TERM`
`sensor_rail`	4.15 A at cold + −30 % vs a 3.0 A polyfuse ceiling	PTC R25 2.0 → 3.0 Ω, solved against the ceiling → 2.88 A	`SENSOR-PTC`

Block	Was	Answer	Tag
`mcu_pdn`	197 mΩ at a wet electrolytic's cold ESR vs 100 mΩ	polymer bulk, ESR **band** 20–50 mΩ → 60 / 92 mΩ	`PDN-BULK`
`metering_unit_pwm`	0.862 A cold / 0.462 A hot vs a 0.675 A setpoint	closed-loop current control → 0.679 / 0.683 A	`MU-ISENSE`
`negative_pulses`	VIN −0.528 V vs the LM5164's −0.3 V	active clamp → −0.068 V	`NEG-CLAMP`
`injector_boost`	ratio missed 11.6× at both corners	11.6× was this block's *nominal*, never a requirement	—

Three of those were more than part swaps, and they are the ones worth reading.

The metering unit needed an architecture change. Over −40…+125 °C the coil's resistance goes 7.39 → 13.82 Ω and at fixed duty the current follows it, ±30 %. A hysteretic current regulator holds 0.679 / 0.679 / 0.683 A at −40 / 25 / +125 °C. Closed loop only works while the regulator has duty left: at the hot corner the coil can draw at most 0.977 A at 100 % duty, so the setpoint sits at 69 % of the ceiling. That bound is checked, because a regulator out of headroom degrades silently back to open-loop.

The negative clamp needed a FET, for a structural reason. A Schottky's drop at the 15 A pulse 1 delivers decomposes as 0.367 V junction + 0.180 V bulk, and only the bulk term falls when parts are paralleled — four of them still leave 0.375 V. The rating is really a demand for ≤ 20 mΩ effective clamp impedance *including the junction*, which no junction device meets and a 5 mΩ FET clears four times over. Still open but now bounded: the comparator's propagation delay is unmodelled, so the question went from *"is a 2 ms excursion acceptable"* to *"is a sub-microsecond one"*.

The PDN requirement had to be a band, not a ceiling. Bulk ESR is also what damps the regulator-against-bulk resonance, so at 20 mΩ the peak climbs back to 92 mΩ. "ESR ≤ 50 mΩ" would have been satisfied by a 5 mΩ part that fails the block.

A near-miss worth recording. The active-clamp branch first shared the battery node with the unprotected one, and at 13 mΩ to ground it shunted the fault for that branch too — reporting −0.177 V where the true Schottky-only figure is −0.528 V, i.e. reporting the problem as already solved. Caught because the clamp diode's own ammeter read 13 mA where 15 A was expected.

Two stale checks were also found. `negative_pulses` still demanded 0.707 J from the TVS, from before the clamp Schottky existed — the TVS no longer conducts on negative pulses at all, and the requirement

moved onto the part that now carries it (13.58 A peak, 14.5 mJ). And a burst-accumulation check conflated first-pulse settling with accumulation; repeats 2–5 are identical to the printed precision.

One block passes in a way worth reading. `emi_filter` was not failed at its corner, because its −40 dB gate is documented in its own final check as being of unknown adequacy pending measured hardware emissions. Failing a corner against an admittedly arbitrary number manufactures a verdict. The X7R −15 % corner is recorded as regression pins with a note that the value sits below the nominal gate.

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2. Circuits not yet built at all

These have no netlist, no schematic and no checks. They are the real gaps.

Missing circuit	Pins	Why it is not built
Negative-input protection for the buck	pin 21 chain	**Found 19 Sep.** The LM5164's VIN is rated −0.3 V to 100 V; the negative limit is not a mirror of the positive one and nothing was sized against it. The TVS clamps correctly to −38 V, which is still two orders of magnitude outside what the buck tolerates. A missing component, not a mis-sized one
~~Intake throttle driver~~	—	**Closed 10 Nov 2026 (U8).** The owner's "throttle" is fuelling to reach and hold 1500 rpm: the firmware's speed governor (`govern.c`), through the injectors and the metering unit. Bosch's drawing gives a throttle valve no pin, so the ECU drives none
~~Pre/post-heat output~~	—	**Closed 10 Nov 2026 (U14), by the owner's rule: follow Bosch.** The MD1CS162 terminal diagram has no heater pin, so the ECU has no heater output. The DSE4522 switches pre/post-heat itself (enabled at 50 °C)

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3. Verification not yet done

The blocks above are each simulated in isolation, at one temperature, against the stimuli named. These are the gaps in the *verification*, not in the circuits.

Gap	Consequence
~~No temperature corner anywhere~~	**Done 19 Sep**, −40/+125 °C across all blocks. Six fail at a corner — `emi_filter` and `can_termination` from *ordinary specified* component behaviour, not stacked worst cases. Being converted from prose into enforced checks
~~ISO 7637-2 negative pulses unsimulated~~	**Done 19 Sep.** The TVS is now modelled as the bidirectional part it is, which is what made the question answerable. Result: the buck's VIN sees −40.7 V against a sourced −0.3 V absolute maximum — see the new gap below
Common-mode EMI unmodelled	`emi_filter` covers differential mode only. Most real automotive failures are common mode, and modelling it honestly needs a layout to give the return path
Reverse-battery transient — now load-bearing	The switch model turns off instantly, so it always looks perfect. That was a caveat; since `negative_pulses` it is a design dependency. If the controller opens fast enough the buck is isolated; if not it sees the full clamped negative voltage. No controller part is chosen and no turn-off delay is sourced, so both bookends are simulated and neither is claimed

Gap	Consequence
Watchdog fault-detection time	`supervisor` bounds only what happens *after* reset asserts, not how long detection takes

4. Documents

Document	State
Reconciled Specification	**Binding.** Rewritten §3 on the fitted controller, §4 gained a hardware fail-safe requirement, register extended to U16
Field Handbook	Current
Sensor Reference	Current
Circuit Reference	Generated from the simulations at build time, so it cannot drift
Memos 01–12	Dated records. Superseded conclusions carry banners rather than being rewritten

Memo 01 and Memo 02 are substantially wrong and say so at the top. Memo 01 misidentified the engine and advised discarding the correct search term. Memo 02 analysed a controller that is not fitted. They are kept because the reasoning that produced them is the record of how the error happened.

5. Unknowns

Closed — 4

#	Question	Answer
U11	Engine identity	`3GK550ETA 4SR1`, app code `GK3.8703`, CPCB IV+ under certificate `P25/2825/24`
U1	ECU connector	Bosch `1 928 405 192 / 194`, **code C** keying
U13	Which controller is fitted	**DSE4522 MKII AMF**, not the KG640C the diagram shows
U12	Does the fuel relay remove power?	**No — it only signals**

Three of these four closed against what this project had reasoned. That is the single most useful thing to know about the desk work: it is productive, and it is not reliable without the machine.

Partly answered — 4

U4 CAN expectations (addresses 234 and 44 confirmed) · **U5** injector drive profile (part confirmed 10 Nov: Bosch 0 445 B76 1X3, CRI-2-16; its electrical data is unpublished, so the drive's numbers are confirmed on the bench) · ~~U8~~ throttle (closed 10 Nov: the governor) · ~~U14~~ pre/post-heat (closed 10 Nov: not on Bosch's drawing; the DSE switches it)

Open — 8

U2 charge-air sensor topology · **U6** pin 09 function · **U7** pins 04/06 sense or feed (no longer blocking: the board uses them as load feeds since 8 Oct) · **U9** catalyst sensor routing · **U10** SENT signal identity · **U15** what `CRS-878` denotes · **U16** alternator rectifier suppressed or not · plus U12's contact-type confirmation, which gates nothing

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6. Decisions open, and blocking

Decision	Blocks	Why it cannot wait
~~Integrated driver IC vs discrete FETs~~	—	**DECIDED 19 Sep 2026, memo 12.** Discrete FETs, keeping the 100 V rail. MC33816 caps at 72 V, L9781 at 80 V (ST data brief, 8 Oct), and both are pre-drivers that need external FETs anyway
~~Passive pulldown vs active clamp~~	—	**DECIDED 19 Sep 2026, memo 12.** Active clamp, built discretely, at every gate. The integrated route relocates this problem rather than removing it
MC33816 last-time-buy 06/08/2027	Fallback path only	Needs a distributor conversation, not another search
~~L9781 datasheet unretrievable~~	—	**CLOSED 8 Oct 2026.** The owner supplied ST's data brief DB3720: `VTank` up to 80 V, a pre-driver for 11 external FETs. The runner-up is confirmed out against the 100 V rail (memo 12 §2.4)
Ideal-diode controller for Q1	`power_input` sheet, `reverse_battery`, `negative_pulses`	Its turn-off time is load-bearing and unsourced. `VBAT_REV_GATE` is undriven on the sheet until a part is chosen
Negative-clamp comparator for Q2	`power_input` sheet, `negative_pulses`	Its propagation delay is the last unmodelled term in the negative-pulse chain. `NCLAMP_GATE` is undriven until a part is chosen
Air-intake shutoff	Nothing electrical	Owners' decision. Nothing in this design can stop an oil-fuelled runaway
Trip-module pickup target	Module ordering	Whether it shares our flywheel target. Sharing is cheaper; separate preserves the independence that is the reason for fitting it

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7. Phases

Phase	Output	State
0	Reconciled specification	Done
1	Research memos	Done — twelve
1.5	Block-level simulation	**Done — 28 blocks, 28 passing** (stop_kill 10 Nov, boost_ovp 12 Nov 2026)
2	**KiCad hierarchical schematic capture**	**Complete — 380 components, 378 with footprints; connectivity OK with 2 named open items (power-ground pin, trip-loop pin)**

Phase	Output	State
3	4-layer PCB layout, DRC, fab outputs	**Prepared for hand routing (14 Oct)**: board with all 379 parts placed by block, classes and rules set, DRC clean, 499 connections to route (decisions log §13). Housing and J1001 are placeholders until measured (brief Group 6)
4	Firmware skeleton	**Core done on the PC, with production logging — 16/16 engine scenarios, 101 unit tests; calibrated from the old ECU's recording (§14). Service link (20 Oct): bootloader, end-of-line record, underspeed warning, laptop tool over USB and CAN — 7 bootloader tests, the tool tested end to end** (decisions log §11, §12, §16). **Simulator (20 Oct): `ecusim`, one program for Windows, macOS and Linux, runs the firmware live or from scripts in a browser; all 16 scenarios pass as scripts** (§17). Next: S32K148 dev kit and the chip layers

Phase 2 — what exists

The project is `hw/ecu25kva.kicad_pro`. Everything in it is generated, and the generators refuse to overwrite a sheet that has content, so Eeschema edits are safe against a re-run.

Sheet	State	Blocks it realises
`power_input`	**Drawn, netlist-checked**	`emi_filter`, `transient_clamp`, `load_dump`, `negative_pulses`, `reverse_battery`
`rails`	**Drawn, netlist-checked**	`buck_preregulator`, `mcu_pdn`, `sensor_rail`
`mcu`	**Drawn, netlist-checked**	`supervisor`, `mcu_pdn`
`injector`	**Drawn, netlist-checked**	`boost_converter`, `injector_boost`, `injector_turnoff`
`metering_egr`	**Half drawn** — metering unit done; EGR blocked on a rail rating	`metering_unit_pwm`, `egr_hbridge`
`sensors_analog`	**Drawn, netlist-checked**	`sensor_ratiometric`, `sensor_differential`, `ntc_frontend`, `battery_sense`
`speed_inputs`	**Drawn, netlist-checked**	`vr_conditioner`, `cam_frontend`
`discrete_io`	**Drawn, netlist-checked**	`discrete_input`, `trip_module_sense`, `relay_driver`
`can`	**Drawn, netlist-checked**	`can_termination`
`connector`	**Drawn, netlist-checked**	— (not a circuit: the harness boundary)

The S32K148 symbol is generated from NXP's own pin table, and no pin number is retyped anywhere in the chain. KiCad 7 ships no S32K symbol. The package pin numbers live only in `S32K148_IO_Signal_Description_Input_Multiplexing.xlsx`, which is not a download but one of eleven workbooks embedded as attachments inside the Reference Manual PDF. That file had been extracted once and not kept, so `docs/pinmap.md` cited a source the repo no longer held. It is now in `refs/`, with checksums, and `hw/extract_pinmux.py` makes the extraction reproducible. Re-extracting confirmed three of pinmap.md's claims independently: 128 port + 16 supply = 144 pins with every package pin accounted for; the six driver-gate pins are all PE=0 PS=0; and exactly four pins on the whole part carry a reset pull, which are precisely the four JTAG/RESET pins pinmap.md reserved.

Three parts have moved from a class to a part number, each against a retrieved datasheet, and all three are in `docs/bom_requirements.md`: `LM5164DDAR` (buck), `TLV76733QWDRBRQ1` (3V3), `TPS3850G33DRCT` (supervisor). Two of them supersede rows in research memo 07, which is now annotated in place.

The buck row is worth reading as a process failure, not just a part change. Memo 07 chose a 60 V part for load-dump margin, correctly, at the time. The input-protection work then raised the TVS standoff and put the pulse 2a clamp at 73.3 V — and `transient_clamp.cir` has carried a *passing* check reading "60 V-class parts are NO LONGER viable downstream" ever since. The contradiction sat between an executable file and a prose table for days, and only the executable one knew. Nothing compares them. It surfaced when someone had to draw the buck and needed a pinout.

Check the netlist, not the plot. Three bugs on the first sheet were invisible on a correctly-plotting page and obvious in the exported netlist: a rotation helper with 90 and 270 swapped, which put the clamp Schottky's cathode on ground; custom part fields starting at property id 2, which KiCad reserves, silently renaming the first one "Footprint"; and labels offset from their wires for looks, which detaches them silently while the page still reads as connected.

A wire endpoint landing mid-span on another wire does not connect — not in KiCad 7's netlister, and not even with an explicit junction element in the file. Interactively, Eeschema *splits* the underlying wire when you drop a junction on it; a generated file has no such split, so the two cross without meeting. The page plots exactly as intended and the netlist reports every tapped part as unconnected. Found on the rails sheet, where the input capacitor, the UVLO divider and the RON resistor all tapped the middle of one wire and all three came back unconnected. `schlib.Rail` now emits one segment per interval and is the only thing allowed to draw a multi-tap rail.

Drawing the sheets is finding things nothing else was going to find. Six so far, all of the same shape — a document and an executable file disagreeing, with nothing that compares them:

- **No watchdog pin.** The pin map assigned 37 signals and none serviced the external watchdog memo 11's supervisor topology requires. A window watchdog nobody feeds resets the board on a timer. `PTE0` / `PTE1` added.
- **The buck was a 60 V part on a 73.3 V rail.** `transient_clamp.cir` had carried a *passing* check saying 60 V-class parts were no longer viable for days, while memo 07's BOM still named one.

- **The trip sense was a GPIO on a three-state loop.** `trip_module_sense.cir` designs and checks three bands with real margin; a GPIO reads two, which discards exactly the distinction the bleed resistor exists to create. Moved `PTD6` → `PTB0` (ADC0_SE4).
- **The cam front-end put 5 V on a 3.3 V pin.** Spec §4 and pinmap §1.3 both gave pin 46's front-end as "pull-up to `5V_SENSOR` → timer capture", and no `sim/blocks/*.cir` file simulated a cam front-end at all — `vr_conditioner.cir` covers the crank and there was no cam equivalent, so the swing was never checked. `cam_frontend.cir` was written, and one 16 kΩ resistor turns the pull-up into the same 10 k/16 k divider the ratiometric channels already use.
- **A ground-referenced kill clamp cannot service a high-side gate.** Spec §4's fail-safe list names pins 03/05 alongside the four low-side gates, and `supervisor.cir` sized a ground-referenced clamp; pinmap §2 confirmed the reset state of all six pins against NXP's own table. All three agree, and all three are silent on the fact that the source of a high-side N-FET swings to the boost rail. The injector sheet puts the 470 Ω across gate and *source*, and moves the kill path into the driver's shutdown input.
- **The power ground has no connector pin.** `GP3.8703.C4.pdf` shows pin 21 as the battery positive feed and no battery negative anywhere, because it is a field troubleshooting diagram and says so: it omits "power grounds, unused pins and internal-only pins". Every other sheet returns to `GND` — the injector low sides put 18 A into it — and at the harness boundary that net has nowhere to go. It cannot be solved by choosing one of the 50 unknown pins: a wrong ground pin is a harness that has to be remade, not a rework.

A seventh finding came out of the parts, not the pins. Every clamp earlier in this project was chosen by forward drop. The crank input's four clamp diodes have to be chosen by *reverse leakage* instead: Schottky leakage at 125 °C puts ~0.90 V of offset across the comparator input's 4.489 kΩ Thévenin, which is 4.5× the whole ±198 mV hysteresis window — the comparator latches and a running engine has no speed signal. `vr_conditioner.cir` cannot see this: its comparator is an ideal switch with no input network, so no clamp, no leakage and no offset exist in it. Tagged `VR-CLAMP` in `docs/bom_requirements.md`.

The battery rail's rating is now enforced, not narrated. VBAT_PROT reaches 40.0 V for 400 ms on a load dump and 73.3 V for ~50 µs on pulse 2a, and `transient_clamp.cir` checks what that excludes: a 40 V-abs-max integrated driver by 1.8×, with the dump alone leaving it no margin. The practical consequence is that integrated automotive H-

bridges and smart switches — a 40–45 V class — are excluded from everything battery-connected here, which is what blocked the EGR half of `metering_egr`.

Two open items were closed by drawing, not by deciding. The injector current-sense placement (`pinmap` §1.6, INFERRED per-bank, "settle when the sheet is drawn") is settled: low-side, ground-referenced, one per bank, because cylinders 1 and 3 fire 240 crank degrees apart and never overlap. And the boost reservoir's missing bleed path — which `injector_turnoff.cir` asked for by name — is now modelled in `boost_converter.cir` as arrangement C and drawn as a 22 kΩ resistor: one hold-cutoff event clears in 21.4 ms against the 26.67 ms between cylinders, for 455 mW burned continuously.

The root sheet is wired and the project has a netlist for the first time. Until 21 September 2026 `hw/ecu25kva.kicad_sch`'s ten sheet symbols carried **no hierarchical pins**, which meant every sheet's own netlist was correct and the project had none: a name on two sheets was two nets and nothing compared them. Each symbol now carries one pin per net its child exports — 162 pins — with a stub and a label, and two labels with the same name are one net.

Exporting it immediately found the second half of the same problem: **`kicad-cli` reported annotation errors**, because each sheet generator allocated `R1`, `C1`, `U1` from its own counter. Ten sheets, ten parts called `R1`. `schlib.REF_BASE` now gives each sheet its own hundred — KiCad's own sheet-number × 100 convention, applied at generation time rather than by a hand pass in Eeschema.

components	**201**, every reference unique
named nets with ≥2 pins	**108** (the four `5V_MAIN` islands merged into one)
named nets with 1 pin	**18**, each a recorded open item
sheet paths	10
`GND` / `3V3_MCU` / `5V_MAIN` / `VBAT_PROT`	139 / 36 / 28 / 12 pins, all spanning the project

Exporting it for the first time found three real defects that ten clean per-sheet netlists could not.

1. **`5V_MAIN` was four nets.** The buck's 5 V output is a board rail — the CAN transceivers' VCC, the five difference amplifiers and the discrete inputs' pull-ups all run off it — and it was a *local* label on every one of those sheets. A local label stops at its sheet's edge, so three of the four copies had no regulator on them, and each sheet's own netlist was perfectly correct. `rails` now exports it.
2. **`3V3_MCU` was three.** Same mechanism, on `can` (both transceivers' VIO) and `speed_inputs` (the comparator's supply, both upper clamp diodes and the bias divider — five pins and no rail).

3. **Five supply pins had no decoupling capacitor.** The difference amplifiers on `sensors_analog` were the only undecoupled supply pins on the board. No simulation block was ever going to say so: `sensor_differential.cir` models an ideal amplifier with no supply pin at all.

All three are now permanent checks rather than one-off observations: one walks every `power_in` pin and asks whether a capacitor sits on its net, and one fails any sheet-local net whose name shadows a project-wide one — the exact shape of defects 1 and 2.

`hw/check_netlist.py` is the gate, and it is the counterpart to `sim/run_sim.py`: that one guards the component *values*, this one guards the *connectivity*. It enforces the same rule — **an exemption is not a pass**. Every single-pin net must be listed with its reason, and a net that stops being single-pin must be removed from the list or the check fails. An open item that quietly closes is as much a drift as one that quietly opens.

The full schematic set plots to `docs/pdf/ECU-Schematics.pdf`, 11 pages.

Phase 2 is done. What remains under these sheets is not schematic work — it is eighteen named open ends, and every one waits on a datasheet, a part number, or the machine:

- **nine** driver / current-sense-amplifier interfaces (requirements drawn, parts not chosen)
- **two** controller gates on `power_input`
- **five** for the EGR bridge, blocked on a gate driver above 73.3 V
- **two** with no part and no pin: `BARO`, and `TRIP_LOOP` — which needs one of the 50 unconfirmed connector pins, as does `GND`

Correction, 21 September 2026 — the connector's part number was in the repo and the sheet said it was not. `hw/connector.kicad_sch` was drawn from `refs/ecu-pinout-extracted.md`, which predates the field visit and still says "a complete pinout requires the connector part number and the OEM pin list". `refs/field-evidence-2026-09-18.md` §2b closed U1 three days earlier, off a photograph of the housing angled to the light: **Bosch `1 928 405 192` / `1 928 405 194`, code C**, `>PA66-GF50<`, EDC17 family — read off the physical part, which beats a catalogue match because it cannot be a mis-identification. **Code C is load-bearing:** Bosch codes these housings mechanically so differently-coded plugs cannot mate. What is still missing is the OEM *pin list*, a different document, and the board-side land pattern — neither number returns a catalogue hit, so the footprint has to come through a Bosch distributor. Same shape as the six findings above, and this one was mine: reading the older reference and not the newer one.

The last two placeholder symbols are gone, 22 September 2026. The crank comparator is TLV3201-Q1 — the clamp's comparator, which meets every line of `VR-COMP`. The five difference amplifiers are **INA592** at their native G = ½: the requirement was ≥ 60 dB CMRR and the drawn discrete network could only reach ~48 dB; the on-die network gives 88 dB minimum. Putting that real figure into `sensor_differential.cir` in place of an inferred 80 dB raised the differential front-end's advantage from 90× to **394×**, referred to the sensor. INA592 is not AEC-Q100; no qualified G = ½ difference amplifier was found.

Baro drawn, 22 September 2026 — 2 → 1 open end. Spec §7 deviation 2 adds it deliberately; NXP MPXHZ6115A from a retrieved datasheet, on 5V_MAIN so the existing `SENSOR_5V_MON` channel makes it ratiometric. Its datasheet does not state AEC-Q100; Infineon's KP236 is listed qualified but its datasheet would not resolve, and it is a footprint change — so that is recorded, not guessed.

The one open end left is `TRIP_LOOP`, and no part closes it. It needs a connector pin *and* a harness wire — it is a new signal, not on the OEM diagram — which is field-brief task 5.2.

Connector decided, 22 September 2026: the OEM Bosch 94-way. Spec §7 deviation 1 had said the ECU would use functionally-split sealed connectors with an adapter harness; `hw/connector.kicad_sch` drew the OEM 94-way on the board. The owner decided for the connector described in the field evidence — Bosch `1 928 405 192` / `194`, **code C** — so the sheet stands as drawn and the spec's deviation is struck through and marked reversed. No adapter harness. What that makes load-bearing: the board-side land pattern (field-brief task 5.3, via a Bosch distributor), code-C keying, and the loom's own answers to task 5.1 (power ground) and 5.2 (`TRIP_LOOP`).

The EGR bridge is drawn, 22 September 2026 — 7 → 2 open ends. It had been blocked since 20 September on one datasheet: DRV8873-Q1's 40 V VM against a 73.3 V rail. It went discrete — four 100 V FETs, two **AUIRS2184S** — and the decode `EGR-DRIVER` demands comes from the driver's structure, not a logic gate: one input per leg, so each leg's high and low are exclusive by construction and the 270 A state becomes a high-side brake. The same family as the injector drivers, chosen the opposite way for the opposite reason. **Coast is asserted by copper** on a shared, pulled-down shutdown, which costs one MCU pin, `PTB10`.

The negative clamp latched, and a passing check hid it — 22 September 2026, 8 → 7 open ends. Choosing its comparator meant setting real thresholds, and the model's were wrong: engage near +0.08 V, release above +0.12 V. At the end of every negative pulse the returning source drove 1.35 A back through the still-on clamp — 6.7 mV across 5 mΩ, never reaching release — so it **latched on and shorted the source.** The

check measured only the clamp's minimum and said RESOLVED; a recovery check failed on both pulses. It is a low-side ideal diode and now releases above −6.8 mV, below zero even with the comparator's full offset.

Two more numbers on that block were artifacts. With the ~100 ns comparator-and-driver delay in, the default timestep let ngspice commit the hysteretic switch on a rejected trial step: the "−0.068 V" minimum was really **−0.177 V** — still inside the −0.3 V rating by 1.70×, which answers the question this block had carried since 20 September. And the pinned 4.74 A clamp-diode peak and a 15.8 mV "settling" difference on pulse 3a were coarse-step overshoot, confirmed on the unchanged original netlist: 4.114 A fine, equal to the flat top and to (206.5 − 0.4) / 50 Ω.

Choosing the clamp comparator found two under-rated parts and a drawn FET of the wrong polarity, 22 September 2026. `NEG-CLAMP` specified its FET at **40 V** and the backstop Schottky was **60 V** — both with a terminal on `VBAT_PROT`, both sized for the negative pulses they catch, neither checked against pulse 2a driving the *same* node to **+73.3 V**. `transient_clamp.cir` carried "every part on VBAT_PROT must clear 73.3 V" as a principle, not a list; it is now a **table**, every part on the rail with its rating and margin, and a part below the line fails. Both are 100 V class now.

And the reverse-battery FET: the sheet drew an **N-FET** with an undriven controller gate, while `reverse_battery.cir` has always modelled a **P-FET**. The node swings +73.3 V to −47.8 V and no controller IC checked spans it — LM74700-Q1 stops at +65 V, LM5050-1-Q1 at −0.3 V. A P-FET with a passive gate network *is* the block's switch model and has no IC ratings to violate. It gives up ideal-diode hold-up on a *shorted*-input dropout, which the spec does not require; that is on the page.

Current sense chosen, 22 September 2026: INA181A1-Q1 on all three channels, 12 → 9 open ends. One part and one gain — 20 — set for the injector, so the 18 A peak reads 1.80 V and a fault to 32.8 A is still measured rather than clipped. Supplied from 3V3_MCU, which makes its output ADC-safe by construction and removes the need for a clamp. The metering FET was also redrawn: it had been placed at `rot=270` since 20 September, which gives diagonal drain and source wires — the same drawing bug the injector sheet had, fixed there first.

Gate drivers chosen, 21 September 2026: AUIRS2181S on all eight gates, and the open-end count fell from 20 to 12. Its first listed application is common-rail injection, and the deciding property is one a parametric search would never filter for: it has **no cross-conduction interlock**. In a half bridge that is the less safe choice; here the high and low switches are in *series* with the coil and must both be on, so UCC27282-Q1 and

UCC27712-Q1 — which have interlock — could not fire an injector at all. It also clears the two ratings the obvious 120 V part (UCC27211A-Q1) fails: 4.1 V of headroom on its HB absolute maximum, and a −1 V HS DC minimum that the freewheel diode violates on every hold off-time. The kill path moves to `HIN`, the one ground-referenced point in the high-side path.

The drivers needed a rail the board did not have — 10–20 V — and **it is made from the boost rail** so it survives cranking. The LM5164 already on the board was the obvious candidate and is ruled out by its own VIN absolute maximum of 100 V against a ~102 V rail. A discrete follower with a current limit carries the ~3 mA; if the boost stops, every driver drops into UVLO and every gate turns off.

Choosing the second part found an error in the first sheet it touched. The injector high side was drawn on 21 September with the battery feeding the bank node through a plain **diode-OR**, on the argument that it needed no extra MCU pin. Looking for a gate driver for it showed that it is wrong three ways, and only the third is visible from a datasheet:

1. **It cannot regulate hold.** With the battery on a diode the only switch in the loop is the low side, and opening it sends the coil's current into the 100 V rail — a fast decay against −87 V, which is turn-off, not chopping.
2. **It leaves connector pins 03 and 05 permanently live**, so a harness short to ground draws current whenever the battery is connected.
3. **The gate driver cannot be bootstrapped.** A bootstrap capacitor charges when the switch node goes low; a battery diode holds it at 12.8 V, so it never does.

The replacement is what `injector_turnoff.cir` already modelled and spends a paragraph of its header justifying: **two high-side switches per bank** onto a common node, plus `D_fw` from **ground**. Hold chopping moves to the high side, the node swings to −0.7 V each off-time, and the bootstrap charges. It costs two MCU pins — `PTB5` and `PTA17`, both FTM0 channels, so the whole injector stage stays on one timer and its edges stay phase-locked. The pin map's "six gate pins" section is now eight.

Phase 3 started 21 September 2026 — the first part is chosen. `TPS40210QDGQRQ1`, the injector boost controller, and it was chosen on **one spec that runs backwards from the usual**: tOFF(min) = 200 ns max. A boost needs 94.1% duty at the 6 V cranking corner; at 150 kHz that off-time floor allows 97.0%, and at the 2.2 MHz most wide-input boost controllers in this class use, the same 200 ns caps duty at 56% — below even the 86.6% the rail needs at a *nominal* battery. Switching slower is the requirement, and the part that looks less modern is the one that works.

And it hit the 73.3 V rail rating for the third time. VDD absolute maximum is 52 V. Same shape as DRV8873-Q1's 40 V `VM`, which took that part off the EGR bridge — solvable here only because this pin draws 2.5 mA plus gate charge rather than a motor's current, so a series resistor and a zener hold it inside the rating. All of it is checked as executable arithmetic under `boost_converter`, not as prose. *(Corrected 22 September: the clamp as first drawn — 47 Ω and a 43 V zener checked as an ideal 43.0 V source — gives 61.9 V with a real part. It is now 220 Ω and a 36 V BZG03C36-HM3, 46.6 V worst case. See §7's parts entry.)*

The converter stage is drawn — `injector` moved to A1 to hold it — and the loop compensation is the one thing on it marked as a starting point rather than a result, because a current-mode boost's loop depends on the inductor's real DCR and the capacitor's real ESR and is a bench measurement.

Small-signal parts chosen, 22 September 2026. Choosing them found two errors, both in gate drive.

1. **Every kill-FET gate could see 73.3 V.** `GATE_KILL` is pulled to VBAT_PROT on purpose, so the kill path has a supply during a 3V3 brownout. Nothing on the net draws DC, so the net follows the rail up to the pulse-2a peak. That is 3.7× a small-signal FET's ±20 V Vgs rating. A BZX84-C12-Q now clamps the net at 12.7 V. See `GATE-KILL-CLAMP`.
2. **The relay FETs' gates were driven from a 3.3 V pin.** The schematic noted them "fully enhanced at 3.0 V", and no 100 V automotive FET checked is specified below Vgs = 4.5 V. An SN74AHCT1G125-Q1 on 5V_MAIN now drives each gate. See `RELAY-GATE`.

Parts chosen: 2N7002BK for the nine kill FETs and the reset inverter, PMV280ENEA for the relay FETs. The same pass fixed a diagonal wire from VBAT_PROT to each relay flyback diode, a drawing bug present since the sheet was first drawn.

Power FETs chosen, 22 September 2026: four part numbers for all sixteen. SQD50P08-25L is the reverse-battery P-FET. IPD90N10S4-06 ×8 covers the negative clamp, metering, the EGR bridge and the injector battery switches. SQM85N15-19 ×5 covers the injector boost-side and low-side switches. DMN15H310SK3 is the boost switch, the one part that needs its Rds(on) specified at a 4 V gate. Two things the choice changed:

- **The symbols.** Every FET had been drawn as `Q_NMOS_GSD`, which is the SOT-23 pin order. DPAK and D²PAK are G-D-S with the drain on the tab. With the TO-252 footprint on the old symbol, the source would have gone to the tab. The power FETs now use `Q_NMOS_GDS` and `Q_PMOS_GDS`, which have the same pin positions and different numbers. The netlist shows every drain on pin 2.

- **Checks that had used placeholders.** The bootstrap droop assumed 60 nC; the chosen part's maximum is 120 nC, which doubles the droop to 126 mV (1.1% of the bootstrap supply). The boost's cranking duty had ignored the switch; with its drop at a hot junction it needs 94.8% of the 97.0% available. The gate pulldowns were sized for Crss = 500 pF; the worst chosen part is 300 pF max.

Diodes chosen, 22 September 2026, with one design error found and one class left open.

- **The boost controller's VDD clamp did not clamp.** It had been checked as an ideal 43.0 V source behind 47 Ω. The real 43 V part is 40–46 V with +0.12 %/K and 45 Ω of impedance, which puts **61.9 V** on a 52 V pin at 125 °C. Its 40 V minimum also conducts on the load dump it was chosen to avoid. It is now 220 Ω and a BZG03C36-HM3: 46.6 V worst case, and 33 mA on the dump. The chosen boost switch's small gate charge is what makes 220 Ω possible. A 39 V zener was tried first, and the `transient_clamp` rail table rejected it at 1.05×. See `BOOST-VDD-CLAMP`.
- **Three stock footprints number their pads against the symbol.** TO-277A (the backstop Schottky) and SOT-23 zeners have the cathode on pad 3. `hw/gen_footprints.py` now derives renumbered copies into `hw/lib/ecu25kva.pretty`. The GATE_KILL zener, footprinted in the previous commit with the stock SOT-23, had been reversed.
- **LOWV-CLAMP is open**, covering the 15 input clamps at 3.0 V and 3.3 V. A sub-5 V zener at tens of µA leaks enough to matter, and the model's sharp-knee zener hides it.
- **Phase-3 notes.** The seven VS-16EDH02HM3 are SMPD, which has no KiCad 7 footprint. The BAV199 pairs are drawn as separate half-parts and need one package per pair.

Transistors and magnetics, 22 September 2026: three more open items, and two of them are design decisions, not part choices.

- **12V_GATE pass transistor.** The DPAK part could not be retrieved. The verified PZTA42-Q in SOT223 would have run 1.01× over its hot rating, so two 3.3 kΩ collector resistors now take half the power. The **105 °C board ambient is an assumption**; the spec states none.
- **POWER-PATH.** The injector and EGR current all flows through pin 21, the EMI filter and a reverse FET sized for 5 A. Pins 04/06 are separate 10 A battery feeds that are used only as senses. Architecture decision needed.
- **BOOST-SLOPE.** By TI's Equation 9, the drawn boost goes subharmonic while cranking (82 mΩ against a 44.9 mΩ ceiling). The 330 µH it needs does not exist at a 2.2 A saturation rating, and a smaller inductor makes the ceiling lower.

- **SENSOR-PTC.** A real PTC is specified from 0.40 to 5.00 Ω, not "3.0 Ω ±30%", and its hold current halves at 85 °C.
- The netlist gate now counts the **power ground** as an open item instead of passing it silently: GND reaches the connector only through sensor-ground pins.

Every chip resistor and capacitor is packaged, 22 September 2026. 188 of them, sized by `hw/assign_passives.py` from the voltage across each part and its continuous power at a 105 °C board. Ratings come from Vishay's CRCW e3 and KEMET's automotive X7R tables, and the netlist gate now fails on any R or C left without a footprint. See `PASSIVE-PACKAGES`. The pass found that 4.7 µF at 100 V does not exist in automotive X7R, so three input capacitors became 2 × 2.2 µF. The EMI model was updated to match.

The 38 parts still without a footprint are all either blocked or chosen separately:

- Blocked on POWER-PATH: L101, L102, F101.
- Blocked on BOOST-SLOPE: L401.
- Blocked on SENSOR-PTC: F201–F203.
- Blocked on LOWV-CLAMP: the 15 zeners at 3.0 and 3.3 V.
- Land pattern needed in phase 3: the seven SMPD diodes, L201 and J1001.
- BAV199 pairs, drawn as separate halves: 5.
- Not yet chosen: C313 (polymer) and C401 (electrolytic). Both chosen 8 October, decisions log §9.

Footprints are the phase-3 input, not a phase-2 gate: 264 of 302 components carry one, and they are exactly the parts with a retrieved datasheet behind them.

POWER-PATH built, 8 October 2026. The injector and EGR loads moved off pin 21. Pin 04 now feeds `VBAT_INJ` (the hold switches and the boost input) and pin 06 feeds `VBAT_EGR` (the EGR bridge). Each has a TVS, an SQM120P10-10m1L reverse P-FET, a Schottky backstop and 4.4 µF of bulk. New block `power_path.cir` (18 checks): EGR stall gives Tj 139 °C at a 105 °C board (155 °C while cranking), and the injector gives Tj 117 °C. The netlist gate checks the split. The result is 333 components, 314 with footprints, and 208 passives packaged. L101, L102 and F101 are unblocked and now need pin 21's reduced load stated before they are sized. Conducted emissions on the two new rails are open. This was done on macOS with KiCad 10 for `kicad-cli` and KiCad 7.0.11's symbol libraries for generation, and ngspice 42.

Pin 21's filter and fuse chosen, 8 October 2026. Pin 21's load is now stated as an upper bound of 2.95 A: the LM5164's 1.75 A peak limit caps the buck's input current, plus the relays, the metering freewheel and the boost VDD feed. The parts are L101 Würth 7447709220 (22 µH, Isat 6.5 A), L102 Würth 744314047 (0.47 µH), and F101 Littelfuse 0437005.WRA (5 A, 63 V, 3.16 A at 105 °C). F101 used to be 20 A. The EMI model now carries the inductors' RDC and self-resonance, and its 150 kHz margin fell to 0.56 dB. The fuse's pulse I^2t is now computed from the simulated waveforms: 16 % of melting on pulse 1, against a 20 % rule of thumb whose pulse-cycling data is still open. 317 of 333 components now have footprints.

SENSOR-PTC built, 8 October 2026. The three PTCs (F201–F203) are gone. Each sensor group now has a TPS7B4253-Q1 tracker following 5V_MAIN, all behind one TPS48110-Q1 overvoltage cut-off. The controller was chosen over the LTC4368 for AEC-Q100: decisions log §7, asked and explained in plain terms. `sensor_rail.cir` was rewritten with pulse 2a through the real pin-21 filter. The trackers' input peaks at 22.3 V (limit 45 V), because the controller's own gate network makes the FET a source follower. The block also found that the clamp node reaches 76.6 V with an unsaturated filter inductor, against 73.3 V in `transient_clamp.cir`; that is now a check. The result is 353 components, 335 with footprints, and 225 passives packaged. Open: the 47 µF hold-up part, and two land patterns.

BOOST-SLOPE built, 8 October 2026: all three approved redesigns are now in. The TPS40210-Q1 runs from its own output through a 12V_GATE-style follower (34.3–45.0 V); the battery only starts it, through 180 Ω and a Schottky. L401 is 150 µH, Würth 7447709151, not the approved 100 µH. Two questions to the owner settled that (decisions log §8): the owner's 10 × 10 × 4 mm suggestion saturates at 1.4 A, and the 12 × 12 mm 150 µH beats 100 µH on stability and saturation. The sense resistor is 62 mΩ. Equation 9 holds at the worst corner by 1.12×. Building it found that the oscillator's −20 % frequency corner would have failed the 100 µH design outright. The result is 362 components, 345 with footprints, and 230 passives packaged.

The three non-ceramic capacitors chosen, 8 October 2026 (decisions log §9). Each is a polymer type, because a polymer's ESR barely moves in the cold. C313 is a KEMET T599 150 µF with a 22 mΩ resistor in series. No datasheet gives an ESR minimum, so the resistor is `PDN-BULK`'s floor. C211 is a Chemi-Con HXC 47 µF 63 V. C401 is now two HXC 82 µF 80 V cans in series, because no part rated above 100 V publishes a cold ESR; the owner chose that over one 160 V wet can. The injector still reaches 18 A in 45 µs at −40 °C, aged. The sims run the 41 µF stack, and every

check uses its worst-case low point, 85.1 V. The 22 kΩ bleed is split 2 × 11 kΩ to balance the cans. The result is 365 components, 351 with footprints, and 232 passives packaged.

Four land patterns drawn, 8 October 2026. Every chosen part with no KiCad land pattern now has one, drawn in `hw/gen_footprints.py` from the maker's recommended layout: - the SMPD for the seven VS-16EDH02HM3 (Vishay 95604; the tab is the cathode, so it is pad 1, matching the symbol's K); - WE-PD 1050 for L201; - TI DGX0019A for the TPS48110-Q1, position 16 empty as on the part; - TI DDA0008J for the three TPS7B4253-Q1. Its thermal pad has a 2.6 × 3.2 solder-mask-defined opening, which no stock variant has.

All four parse and plot in kicad-cli. They are written in KiCad 7 format, because the installed KiCad 10 library is newer than the project files. 363 of 365 components now have a footprint. The two left are J1001, which needs the Bosch drawing, and D806, the trip-loop clamp.

A filter on each new supply feed, 8 October 2026 (decisions log §10). New block `feed_emi.cir` measures conducted emissions the way CISPR 25 does, through a 5 µH / 50 Ω LISN. As drawn, the injector feed was 33 dB over Class 5 and the EGR feed 48 dB over, because POWER-PATH had taken both off pin 21's filter. Each feed now has a Coilcraft XAL1510 10 µH inductor, two 82 µF HXC bulk cans and two R-C dampers (0.68 Ω + HXC). Margins are 17.7 dB (injector) and 9.3 dB (EGR) under the limit. The dampers keep the filter stable against the constant-power loads, 1.47× with aged capacitors. The filter also cuts pulse 2a at the loads to 28.7 V. It sets a firmware requirement: hold chopping and EGR PWM at or below 25 kHz. The result is 379 components, 377 with footprints, 236 passives packaged, and 26 of 26 blocks passing.

Firmware, proven on the PC, 11 October 2026 (decisions log §11). `firmware/core/` is portable C. It covers crank/cam decoding, injection timing, the speed governor with droop, rail pressure, sensor faults, latched shutdown and J1939 to the DSE4522. `firmware/sim/` runs it against a Python model of the engine, generator and rail. All eleven spec scenarios pass (`.venv/bin/python firmware/sim/run_sil.py`, 36 s), and so do 46 unit tests (`make -C firmware/sim test`). A mid-build expert review found 2 critical and 7 important defects, among them a start that peaked 3 rpm under the overspeed trip. All nine are fixed, each with a test that failed first. The watchdog kick period is now fixed at 10 ms, so the TPS3850 window straps (SET0/SET1, CWD) can be chosen to accept 5–20 ms. A final review found that the firmware spec had assumed a normally-open metering valve, which contradicts the trip module's design. The firmware now assumes normally closed. **The real valve must be identified**: if it is normally open, the trip module's contacts have to move.

Production logging, 11 October 2026 (decisions log §12). Fault memory with snapshots, a 64-entry event log, lifetime counters (engine hours, starts, trips, watchdog resets, time at load) and a trip recorder now survive power-off. They live in a 3 KB image saved to two alternating copies with CRC-32. DM2, DM3, DM4 and engine hours are read out over J1939. The watchdog now pulses only while the main loop is alive. The ms counter wraps hourly, a lost tooth logs as a crank fault, and a too-long fault message is never half-sent. There are now 15 scenarios and 78 unit tests, all passing.

Field data, 13 October 2026 (`refs/field-evidence-2026-10-13.md`). Jeevan sent: - **Firing order 1-2-3**, which confirms the firmware assumption. - **Governor class ISO 8528-5 G2**, which becomes a requirement. The simulated governor meets every G2 limit. On the real engine, though, the load-rejection peak must stay under our own 1650 rpm trip, which is tighter than G2's +12 %. - **The KOEL injector number** `4H.508.70.0.PR.REV-1` (U5). - **Photos of the old ECU**: Bosch `0 281 021 133` , type `MD1CS162` , KOEL `T4.435.01.0` . Its connector face shows **6 large pins (1–6) and 4 rows of 22**, so the J1001 placeholder is corrected. A second, unidentified ECU is also recorded. - **An engine-off CAN trace.** It holds only UDS diagnostics (0x7E0/0x7E8), so **U4 stays open.** Its ECU names itself `KOEL_TC_82.5KVA` / `4R1190ETA4G1` , a 4-cylinder 82.5 kVA set, not ours; logged as **U17**.

The September DSE record also shows that the DSE shuts down only on the DM1 **Protect** lamp, which our firmware never sets. Fixed on 14 October with the owner's go-ahead: a shutdown fault (overspeed, crank loss) now sets **red and Protect** in DM1; a sensor fault sets amber only. Decisions log §12, addendum.

PCB prepared for hand routing, 14 October 2026 (decisions log §13). `hw/ecu25kva.kicad_pcb` holds all 379 parts placed by block, with: - the 4-layer stack-up - ten net classes, with widths computed from current - a custom rule for pads inside a part

`hw/check_pcb.py` passes: footprints, nets, classes, widths, and DRC with no clearance or courtyard errors. 499 connections are left to route.

The outline, holes, cooling band and J1001 are **placeholders**; the inspection brief's new Group 6 asks for the measurements. **The parts don't fit one side of the spec's 160 × 120 mm guess**, so the placeholder is 180 × 130 mm. If the old board is smaller, parts go on both sides. The routing guide is `docs/layout-guide.md` .

The old ECU's own measurements, engine running, 14 October 2026 (`refs/field-evidence-2026-10-14.md`). This is a 35-minute calibration recording (XCP, MDF 3.0) from a 3-cylinder engine at 1500 rpm on a 12 V system, with no faults stored. It is not a CAN bus trace, so **U4 stays open**. It contradicts five firmware assumptions: - **The metering valve**

behaves as normally open. Current falls as pressure rises, and rises to stop the engine. Firmware assumes normally closed. This is a safety change: the trip-module contacts have to move. Confirm with the valve's part number. - **Injection timing** is 11.5–15° before TDC. Firmware uses 5°. - **Rail pressure** follows load, 827–1057 bar. Firmware uses a fixed 1000 bar. - **Injector constants:** dead time about 330 µs and flow constant about 0.0020. Firmware uses 300 µs and 0.0009, so it would hold each injector open about twice as long as needed. - **The governor is isochronous,** with no droop. Firmware applies 4 %.

Two more readings: - The OEM governor dips **9.8 %** on a load step, at G2's limit. Our simulation is likely optimistic. - The intake-air-temperature input reads 0 mV, so no sensor is fitted (U2).

Applied the same day with the owner's go-ahead (decisions log §14): normally-open valve, measured injector law, and timing and rail tables. 80 unit tests and 15/15 scenarios pass. **Still open: the trip module has to cut the injector feed, pin 04**, not the valve. That is written up in spec §3 item 6, with the field survey in brief Group 7. It is external wiring, and not yet done.

The Bosch pinout becomes the master, 7–8 November 2026 (decisions log §18). The owner sent Bosch's terminal diagram for the MD1CS162 (`refs/T4.435.04.0.PR-terminal-diagram-MD1CS162.pdf`, pin list `refs/bosch-pinout-md1cs162.md`). The owner's ruling: we replicate this ECU, the drawing applies to our engine, and we use only Bosch pins. It disagrees with KOEL's harness diagram on power: - **No permanent battery feed.** Pin 71 (ignition) closes the main relay in hardware, through a new wake FET on pin 50. The relay feeds pins 04/06, the board's only supply. The electronics chain moved from pin 21 to pin 04. - **Pin 21 is the water-in-fuel input.** Pins 01/02 are the power ground, which closes the gap open since 19 September. Pin 09 is the GCU's analog sync input. Pin 14 is the diagnosis request and pin 25 the stop switch. - **A latch bug found and fixed.** Pin 50's flyback diode to the board's supply would have fed the board through the always-live relay coil and held the relay in for good (22.8 mA, 2.7 times its release current). It is now a TVS to ground. `sim/blocks/main_relay.cir` shows both. - **The trip loop is gone.** It had no Bosch pin. The trip module now opens the main relay's coil feed (spec §3 item 7). - **Firmware, ABI 2:** - After-run: the relay is held until the engine has stopped and the fault memory is written, at most 60 s. - Pin 25 latches SHUTDOWN, like the ignition. **Since 10 Nov it also kills the gates in hardware**, for when the firmware is what's misbehaving (decisions log §18). - Water in fuel raises an amber SPN 97. - The service link runs at 1000 kbit/s, Bosch's rate for 87/86. - **The netlist check had been passing its decoupling check on nothing** under KiCad 10. It now parses KiCad 10's netlist and checks 50 supply pins.

Every check passes on this Windows machine: 26/26 circuit blocks; the netlist with **0 open items**; the firmware unit tests, the boot tests, 16/16 scenarios and the service tool; and the board check, schematic parity included. KiCad 10.0.6 drops every pin on an auto-named net from parity, so since 9 Nov `check_pcb.py` checks those pins against the exported netlist (HANDOFF §2). **The Rust simulator is updated but unbuilt**: there was no Rust toolchain here.

The prototype board was brought up to the Bosch pinout on 9 November (decisions log §18, "The prototype, brought up to it"): - 30 new parts, each input's front end placed at J1007. - The three new inputs on PD8–PD10. - 0 DRC errors, nothing unconnected, and the fab files regenerated.

The boost cannot run away, 12 November 2026 (decisions log §19). The owner asked for fixes for the 100 V boost and for heat: - **A hardware overvoltage cutoff.** A 110 V zener (D417) drives the boost controller's DIS/EN pin, so an open feedback resistor can no longer push the rail into its 150 V parts. It trips at 105.0–119.3 V over both tolerance corners (`sim/blocks/boost_ovp.cir`). A 5.1 V clamp protects the pin, and `BOOST_DIS` (PTD15 on production, PD14 on the prototype) lets the MCU stop the boost too. Five new parts on each board; `check_netlist.py` checks the circuit. - **Firmware, ABI 3:** above 106 V on the rail, it stops the boost and shuts down (red DTC, SPN 520192). It also stops the boost while shut down. An **EGR stall guard** lets the bridge coast after 1.5 s over 5 A (amber DTC, SPN 2791 FMI 6), and holds it off after the third stall. - **Coating and heat:** the board is conformal-coated before it goes on the engine. Bring-up step 7 adds a 10-minute full-load heat check with a 100 °C limit.

The application in Simulink and Stateflow, 13 November 2026 (decisions log §20, `model/README.md`). The owner asked for a state-flow model that generates the code the ECU runs: - **Twelve models**, built by scripts in `model/` : six Stateflow charts (the engine state machine, and the boost, EGR stall, water-in-fuel, underspeed and main-relay guards) and six Simulink models (governor, rail control, the two tables, injector law, trip decision). Constants are read from the firmware's C. - **Embedded Coder generates the C** into `firmware/gen/` . Building with `ECU_MODEL` swaps it in behind the same functions. - **Proven equal:** all unit tests and 16/16 scenarios pass on the generated build; the back-to-back test agrees exactly on millions of random inputs (the tables to a unit in the last place). - `ecu_app` : the whole application closed loop with an engine, simulated in Simulink. - **Flashing:** the code goes in the firmware image like any other. The STM32 chip layer is still the missing piece, for either build. - **14 Nov:** boost-rail warnings (held high: the

zener clamping; low while fuelling: a dead boost), the state machine owning the ignition cycle, and `./verify.sh` to run every check at once (decisions log §20).

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8. Work at the machine

Ordered by what it unblocks.

Re-ordered 21 September 2026, after the schematics were drawn. Two items moved to the top that were not on this list before, because they came out of capture rather than out of a document. The full tasks are `docs/research/08-engine-inspection-brief.md` **Group 5**, written for whoever is at the machine.

Task	Closes	Unblocks
~~Continuity from every populated cavity to chassis — where is the power ground?~~	—	**Closed 8 Nov 2026 by Bosch's drawing: pins 01/02.** A continuity check still confirms it
~~Cavity census~~	—	~~A pin for `TRIP_L00P`~~ The loop is gone (no Bosch pin)
~~Whether the main relay has its own suppressor~~	—	**Closed 10 Nov:** pin 50 drives the main relay and 69 the buzzer relay (owner). Pin 50's TVS, and pin 69's flyback diode to the board's supply (fed, like the buzzer coil, from the main-relay line), are safe with or without a suppressor in the relay. Settled by the owner on 10 Nov: pin 25's polarity as Bosch draws it; pins 09 and 14 as analog and digital inputs, functions configured later
~~Pin 21 with the ignition off~~	—	**Closed 10 Nov 2026 by the owner:** pin 21 is the water-in-fuel sensor, as Bosch draws it
~~Trace who switches the heater feed~~	~~U14~~	Closed 10 Nov 2026: not on Bosch's drawing, so not the ECU's
~~Follow the throttle's 6-pin cable to where it ends~~	~~U8~~	Closed 10 Nov 2026: speed is governed by fuel quantity; no throttle driver
~~Injector part number from the body~~	**U5**	**Part confirmed 10 Nov (owner):** Bosch 0 445 B76 1X3 (CRI-2-16), as on the drawing. Its coil data is not published, so the 100 V boost and 18 A peak stay the CRI-2 family's and are measured on the bench with a real injector (bring-up step 4); the setpoints are firmware values
~~Alternator rectifier part number~~	~~U16~~	Closed 9 Oct 2026: designed for the suppressed load dump, no field check needed
Resistance across the charge-air element, cold	**U2**	The `ntc_frontend` populate choice
Crank VR output while cranking, cold *(supervised)*	—	Confirms the ±198 mV hysteresis window against the real sensor, not a 2 V estimate
Contact pitch and row spacing, with calipers	—	Confirms which Bosch family drawing is the right one for the board-side footprint
Relay contact-type check	U12 confirmation	Nothing — recorded for completeness

Nothing on this list can grow the board any more: U8 (throttle) and U14 (heater) closed on 10 Nov, because neither is on Bosch's drawing. Everything left changes a value or closes a register entry.

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9. Resume here — next session

Updated 8 November 2026, after the Bosch pinout (decisions log §18). Every check passes: - 26/26 circuit simulations - the netlist gate, with **0 open items** - the firmware unit tests, 7 bootloader tests, 16/16 engine scenarios and the service tool end to end - the PCB rules and generator self-tests - the board check, including the schematic links and parity (on Windows, pins on auto-named nets are checked against the exported netlist: HANDOFF §2) - all 26 prototype tests (9 Nov)

On Windows these all run with KiCad 10's bundled Python and ngspice, plus `zig cc` (pip `ziglang`) for the firmware. See HANDOFF §2.

Where it stands

- **Schematic:** complete, 405 parts, **on Bosch's pinout** since 8 Nov (decisions log §18). Regenerating it is deterministic.
- **PCB:** prepared for hand routing, not routed (decisions log §13).
- **Prototype board (17 October; Bosch pinout 9 November):** routed and fab-ready in `prototype/` : the full production circuit with an STM32G474, screw terminals, 300 × 200 mm, 4 layers. 0 DRC errors, nothing unconnected, schematic parity clean. Gerbers, drill, positions and BOM in `prototype/fab/` ; bring-up guide `prototype/docs/bring-up.md` (decisions log §15). **Reviewed 19 October:** the review found the watchdog kill path lost by the 15 October supervisor fix. D302 (BAT54Q) restores it, in production too. The 5 V buck's output and switch node now run on the outer layers, sized for its 1 A. The pour checks were made able to fail.
- **Firmware:** runs on the PC, calibrated from the old ECU's own recording (§14). **Service link (20 October):** a bootloader with the image header written last, the end-of-line record with its analog trims, an underspeed warning (SPN 190 FMI 18, amber, no shutdown), and a laptop tool (`firmware/tools/ecu_service.py`) over USB or CAN channel 2, all proven on the PC (decisions log §16, firmware guide §16). The chip layers (flash, CAN, USB, reset) are still to write.
- **Simulator (20 October):** `simulator/` , mode 1 (firmware only). `ecusim` is a single Rust program. It runs the unchanged firmware against the engine model, live or from `.seq` scripts, at 1× to about 100×, and traces everything in a browser. Firmware builds load from the UI. The 16 SIL scenarios pass as scripts. Guide `simulator/README.md` ; decisions log §17. Modes 2 (board fed over USB) and 3 (bench rig) are next.

- **Safety change written up, not yet wired:** the overspeed trip opens the main relay's coil feed (spec §3 item 7, which supersedes item 6's pin 04).
- **Stale since 8 Nov:** the Rust simulator's build (its source is updated to ABI 3).
- **The prototype is on the Bosch pinout as of 9 Nov.** Placement and routing were updated in place; decisions log §18.

Waiting on the machine (inspection brief)

1. ~~7.1, the metering valve's part number.~~ **Confirmed 10 Nov (owner):** Bosch 0 928 400 820 (CB18), as on the drawing. Bosch's 0 928 400 metering units are normally open, as the firmware and the stop design assume.
2. **Group 6, the old ECU's measurements:** housing, board, and the J1001 header. **Deferred (owner, 10 Nov 2026):** the prototype is tested physically first, and the production board is changed and fitted to the housing only after that. It blocks the production board, not the prototype.
3. ~~5.1 / 6.4, which pins are battery ground.~~ Bosch's drawing: 01/02. Pin 25 follows Bosch's drawing (owner, 10 Nov). Pins 09 and 14 are settled as inputs; their functions are configured later.
4. **A running-engine J1939 trace on the DSE's CAN** (U4). The 14 Oct file was an XCP recording, not a bus trace.
5. **7.2–7.4, any existing trip module and the main relay's coil feed:** needed when the trip is fitted.
6. **Emissions certification (CPCB IV+, certificate `P25/2825/24`).** It was granted with Bosch's ECU and calibration, so a replacement ECU will probably need it re-checked. **Deferred (owner, 12 Nov 2026):** the ECU is proven working on the prototype first. The certification is taken up before any unit goes into a customer's genset. It doesn't block the prototype.

Needs a person

- **Routing the PCB:** `docs/layout-guide.md`, once Group 6 replaces the placeholders.
- **MC33816AE last-time-buy:** a distributor question (fallback path only).

Next that needs neither

- **Build the Rust simulator** at ABI 3 (`cargo test`), and rebuild the committed Windows `.exe`.
- **Quote and build the prototype:** 2 oz outer copper, 300 × 200 mm, 95 pin necks at 0.15 mm (decisions log §15, "The cost to know before quoting").
- **The S32K148 port:** running the firmware core on NXP's evaluation board, with the engine signals simulated on the bench. It needs the board, a debugger and a toolchain on this machine.